Sprint reviews, retrospectives and the feedback action log

> **Citations to files that are not in this tree.** An audit found 39 paths cited across 19 documents that no longer exist — the fourteen-pool model, the v1/v2 generators and the drawing scripts that went with them, and the Sprint 0 Node scaffold. Nothing was lost: all of it is in git history, and every citation is listed with what replaced it in 18-cited-files-no-longer-present.md.

This is the review and feedback record for the two sprints. Sprint goals and item IDs come from `docs/07-product-backlog.md`; acceptance evidence is the file named in `docs/03-test-record.md` and present in `screenshots/` or `diagram/`. No dates, attendee lists or verbatim quotation of stakeholder remarks survive in the workspace, so the document records *what was shown, decided and agreed*. It makes no claim about who said it or when; the gaps are listed under Assumptions.

Read with: `docs/07-product-backlog.md` (goals, owners, Definition of Done), `docs/08-test-plan.md` (test results quoted here), `docs/02-status.md` (what is not done), `docs/04-modelling-decisions.md` §7 (limitations), `docs/05-v2.1-changes.md` (the R2 change set).

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1. Sprint Review 1: Initial Release (R1)

Sprint goal.

> Replace the v1.0 drawing with a measured layout engine and close the two behavioural gaps the gap audit found (exception branches with an owner, and compensation for committed resources). Then prove the whole collaboration runs on a live engine with no incidents.

Increment shown. The v2.0 build: the collaboration, the regenerated forms, the rebuilt workers, and both scripted paths run live against c8run 8.10.0-alpha5.

1.1 Acceptance evidence presented

Item	Acceptance evidence shown	Reviewer outcome
PB-01 gap audit	`docs/01-gap-audit.md`, with the v1.0 measurements from `tools/analyse_layout.py`	Accepted: every rubric descriptor answered with a number or an artefact
PB-02 – PB-04 structure, lanes, messages	Model opened in the Modeler; 15 participants, 14 lanes, 9 executable processes	Accepted
PB-05 layout	`python3 tools/analyse_layout.py` output: sequence-flow overlaps 0 pairs, message-flow overlaps 0 pairs, 0 diagonal px	Accepted, because the two rows the build is gated on are zero

Item	Acceptance evidence shown	Reviewer outcome
PB-06 collapsed sub-processes	3 collapsed sub-processes, steps still in the XML	Accepted
PB-07 – PB-08 exception owners and letter timers	9 error codes, 18 boundary events; `P7D` and `R/PT168H` present	Accepted
PB-09 – PB-11 forms	36 forms on 39 user tasks, candidate group on every task, validation present	Accepted
PB-12 accessibility	Descriptions on some fields; **no grouping headings**	**Not accepted as complete**: recorded as partly done
PB-13 – PB-17 workers	`mvn` build green; subscriptions observed; the three v2.0 job types registered; card-data and audit rules inspected	Accepted
PB-18 compensation	Model wiring shown (2 boundary compensation events, 2 handlers, 2 throws)	Accepted for modelling; **live trigger not shown**
PB-19 capped retry	`TRT_Auto_RecordRetry` and the cap-3 chain	Accepted
PB-20 – PB-22 deployment and paths	Deploy output (9 process definitions, 36 forms); `run-happy-path.log` (11 steps, `incidents: none`); `run-exception-path.log` (`REFERRAL_PACK_UNREADABLE` caught, case joins the missing-information loop, no incident)	Accepted; demonstrated live
PB-23 form rendering	`TOTAL=36 FAILED=0` against form-js; `screenshots/camunda-forms-rendered.png`	Accepted
PB-24 records	`docs/03-test-record.md`, `docs/04-modelling-decisions.md`	Accepted

1.2 What was demonstrated

The normal path was driven end to end: referral message → Medical Secretaries document check → Consultant decision → Outpatient Bookings slot choice → Call Handling telephone contact → attendance and consent → authorised treatment request → Finance funding decision → payment → treatment confirmation → clinic letter drafted, processed and sent. The unreadable-referral exception path was then driven and held at `SEC_EGW_InfoRequest` with no incident. The model was opened in Camunda Modeler 5.51.0 on Camunda 8.10 (alpha) and the layout analyser was run in front of the review.

1.3 What was not finished

Item	State	Stated at the review
PB-12 (grouping half)	Descriptions on some fields only; no group headings emitted	Not delivered; no `group` component exists in any form
PB-18 (live trigger)	Compensation modelled, handlers registered, never fired	Modelled and testable; not demonstrated
PB-30 scope (branch runs)	Declined payment, confirmation lost and the three-month escalation not driven	Not reviewed in this sprint; they were not in the Sprint 1 committed set but were named as the remaining behavioural gap

Item	State	Stated at the review
PB-32 scope (Tasklist)	Tasklist Task pane blank; `401` from `/v2/authentication/me`	Environmental; forms proven by the renderer census instead

1.4 Decisions made

#	Decision	Reasoning
D1	Zero overlaps is prioritised over crossing count	The brief names overlap and parallel adjacency as the fault and crossing only as something to minimise; spreading routes across corridors removes the overlaps but creates crossings
D2	The four external suppliers stay black-box participants	The 4 MB deployment batch ceiling makes a further executable process expensive, and their internals cannot be verified from the case study
D2 (revisited, v14)	**Reversed.** All six outside participants now carry a small message-driven process and a drawn hand-off in each direction	The case study *does* describe the boundary well enough to draw it: slots returned, dispatch result returned, payment status and reference returned, capacity reported. Three of the six pools had no drawn connection at all, which read as a modelling mistake. The batch ceiling is now an explicit, open risk, no longer the reason for the omission: `docs/11-white-box-external-participants.md` §5, `docs/09` R-01
D3	The three embedded sub-processes stay collapsed	No `no-bpmndi` warnings inside a collapsed box; drawing them open was measured against the closed form and deferred
D4	The Tasklist authentication `401` is treated as environmental	All 36 forms render in the renderer Tasklist uses, so the Forms deliverable does not depend on the webapp
D5	Collapsed sub-process steps are not given their own diagram interchange	The warnings they produce are `no-bpmndi`, which is a drawing omission rather than a modelling error

1.5 Questions raised

#	Question	Immediate answer	Followed up as
Q1	How many dashed message lines does the diagram actually carry, and are they all needed?	66 at the time; Camunda 8 does not read a message flow at run time, so the drawing can carry one per pool pair	FB-01 → carried into Sprint 2 as PB-26
Q2	Why did crossings rise from 42 to 148?	The measured price of removing overlaps; A/B will confirm	FB-02
Q3	What are the 30 Modeler warnings?	All one rule, `no-bpmndi`, inside the collapsed sub-processes	FB-04
Q4	Can the forms be proven without Tasklist?	Yes: form-js is the renderer Tasklist uses	FB-05
Q5	Why are there 43 start events when the target was two per pool?	A Camunda 8 message start subscribes to one message name; collapsing them changes the message protocol	FB-06
Q6	Does compensation work?	It is wired and the handlers register; it has not been triggered	FB-07 / PB-31

1.6 Limitations stated at the review

Taken from `docs/04-modelling-decisions.md` §7 and `docs/02-status.md`: crossings rose; four flows clip a boundary event; the canvas is a single very large diagram (exported as SVG and per-pool crops for that reason); the four external suppliers are deterministic stubs; idempotency state is in memory; the deployment batch ceiling constrains the design; three branches are modelled but not run; and Tasklist cannot draw the forms because of the local `401`.

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2. Retrospective 1

What went well	What did not	Improvement action for Sprint 2
The gap audit was measured rather than asserted, so the rebuild had a target list instead of a redraw	The forms `date` field-type defect (DEF-12) and the import-order defect (DEF-13) were found very late, by accident of checking the renderer, not by a planned test	Add the form-js render census to the standing gate set so a field-type defect cannot survive to the end again. Owner D, second E.
Running the model, not reading it, found eleven real structural defects (DEF-01 to DEF-11), several of which would have broken a live demonstration	A drawing-only change in Sprint 2 could silently drop a scored element; nothing checked that a rebuild preserved the previous release	Build the executable preservation audit before any Sprint 2 drawing change (owner E, second B).
The layout gate is objective and cheap: two numbers must be zero before a build is accepted	The crossing regression (42 → 148) was only explained after the review raised it; the trade-off was not written down before the change was accepted	Record a written trade-off (the A/B numbers) in `docs/05-v2.1-changes.md` for any measure that gets worse; owner B, second A.
The two-owner pattern worked: every hand-over in `docs/07-product-backlog.md` §7 was accepted by someone who did not make the change	The three undriven branches were known at the Sprint 1 review and still were not run	Drive the three branches early in Sprint 2, before the compaction work: owner C, second E (see FB-07, not carried out).

Actions agreed at Retrospective 1. Three of the four were completed. The fourth, running the three undemonstrated branches early in Sprint 2, was not carried out; committed compaction work and the preservation gate displaced it. The reason is recorded, and the outstanding impact is that `TC-E2E-03` to `TC-E2E-05` remain `NOT RUN` and risk R-05 stays open. It stays on the list as FB-07.

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3. Sprint Review 2: second release (R2)

Sprint goal.

> Reduce the drawn density of the shipped model without removing a single scored element, and re-prove the behaviour afterwards on the rebuilt file.

Increment shown. The v2.1 rebuild: thinner drawing, same elements, both paths re-run, model re-validated in the Modeler.

3.1 Acceptance evidence presented

Item	Acceptance evidence shown	Reviewer outcome
PB-25 preservation gate	`python3 tools/verify_preservation.py` → **62 checks, 62 passed, 0 failed**	Accepted: the gate ran before the change, as agreed at Retrospective 1
PB-26 message-flow thinning	Message flows 66 → 31; dashed line 421,424 px → 174,820 px (–59%); exception hand-offs kept in preference to routine ones	Accepted
PB-27 end-event merge	End events 60 → 32; `_exception_ends()` walks forward from every boundary event so exception ends stay distinct	Accepted
PB-28 lane consolidation	Lanes 20 → 14; the 15 participants and their names untouched	Accepted
PB-29 Modeler and export	Camunda Modeler 5.51.0 on Camunda 8.10 (alpha), **0 errors**; `tools/validate_model.js` 0 warnings, 0 unresolved references; SVG and PNG regenerated; 15 section crops refreshed	Accepted
Behaviour re-proof	Both paths re-run on the rebuilt file: normal path all steps with `incidents: none`; exception path held at `SEC_EGW_InfoRequest` with `incidents: none`	Accepted
Deployment	9 process definitions at v17 and 36 forms at v18	Accepted
Evidence refresh	`operate-dashboard.png` (21 running instances, 0 with an incident), `operate-instance-completed.png` (executed steps and real variables), `tasklist-open-tasks.png`	Accepted

3.2 What was demonstrated

Both paths from Sprint 1 were re-run on the rebuilt v2.1 file. That is the substantive claim of R2: removing 35 drawn dashed lines and 28 end circles changed only the drawing. The preservation audit and the two clean runs are the evidence. The Modeler was opened live to show the status bar at 0 errors.

3.3 What was not finished

Item	State	Stated at the review
PB-30 three branches	Not started	Still modelled, still not run; the reason is that the committed compaction work and the preservation gate took the sprint
PB-31 live compensation	Not started	Same reason
PB-32 Tasklist authentication	Not started	`401` unresolved; forms proven by the renderer census
PB-33 start-event reduction	Deferred	Requires a message-protocol change; arithmetic recorded in `docs/05-v2.1-changes.md` §4
PB-34 crossing reduction	Deferred	Crossings are 197, up from 147, as the measured cost of merging end events

Item	State	Stated at the review
PB-12 grouping	Still not delivered	Open gap

3.4 Decisions made

#	Decision	Reasoning
D6	Accept the crossing rise (147 → 197) as the price of 28 fewer end circles	A/B measured: pool-wide merge gives 27 ends and 208 crossings; the lane-scoped variant gives 32 ends and 197 crossings and was chosen as the better of the two
D7	Keep the three sub-process boxes collapsed	Drawing them open measured 21 lines through a shape against 4, and added 3 overlaps; the closed form is the better read
D8	Treat the model file as the source of truth and fix the prose that disagreed with it	The shipped file matches `README.md`, `docs/02-status.md`, `docs/03-test-record.md` and `docs/05-v2.1-changes.md`; the drifted document was a configuration-management item, not a model change. **Executed in R3:** it was deleted and replaced by `docs/06-diagram-engineering.md`
D9	Do not bump `versionTag` for this release	Left as a recorded shortcoming; the engine deployment versions identify the release in the meantime
D10	Carry PB-30 to PB-35 into the next cycle, with no attempt to compress them into the remaining time	They are Should or Could items; the Must items of both releases are delivered

3.5 Questions raised

#	Question	Immediate answer	Followed up as
Q7	If 30 Modeler warnings remain, is the model valid?	Yes: all 30 are `no-bpmndi` inside the collapsed sub-processes; nothing is reported as an error, and the parser gate is clean	FB-04 closed
Q8	Does the thinner drawing still run?	Both paths re-run with `incidents: none`; Camunda 8 does not read a message flow at run time	Closed
Q9	Why is the worker README table short of job types?	The three v2.0 types were added to the code and the model but not to the table, whose summary line is also stale	FB-09
Q10	Is the version tag right for this release?	No, it still reads 2.0.0	FB-10
Q11	Which document describes the shipped diagram?	The model file; `docs/06` described a further pass that was not in it. R3 replaced that document	FB-11 (closed)
Q12	Where is the version-control history?	There was no repository in the workspace at the time of the review; one was created afterwards	FB-12 (closed)

3.6 Limitations stated at the review

Everything from Sprint Review 1 §1.6 still applies, plus two reduction targets that remain unmet (43 start events, 197 crossings), the four clipped lines and the 30 accepted Modeler warnings. The three undriven branches, the un-triggered compensation, the Tasklist gap and the documentation drift items are unchanged. The whole model remains a local, single-instance demonstration over simulated external services.

4. Retrospective 2

What went well	What did not	Improvement action for the next cycle
The preservation gate was built first, exactly as agreed at Retrospective 1, and it caught the question of whether a drawing change had touched anything scored	The documentation drifted from the artefact: one document described a diagram state that was never shipped, the worker README table was three job types short, and a cited screenshot does not exist	Correct the drift items before submission. Owner E, second A. R3 closed three of the four: the drifted diagram document was deleted and replaced by `docs/06-diagram-engineering.md`, the worker catalogue was completed, and the drawing itself was re-engineered (197 → 94 crossings, 4 → 0 clipped lines). Two readability changes followed, one end event per terminating path (32 → 65, none merged) and dashed lines cut from 31 to 15. The missing screenshot (DEF-17) stays open
The compaction was measured before and after, and the one regression (crossings) was confirmed by A/B	The three undriven branches were known for two sprints and still were not run	Schedule the branch runs as the first item of the next cycle, before any drawing work; owner C, second E.
Re-opening the model in the Modeler cleared the open question from the gap audit, and the parser gate now reports 0 warnings and 0 unresolved references	No version-control repository exists, so the baseline depends on folder copies	Initialise a repository and commit the released state as the baseline (owner E, second A).
Both paths were re-run on the rebuilt file, so R2 is backed by behaviour and not only by measurement	Accessibility remains partly delivered: no grouping headings and no WCAG check	Emit `group` components in `tools/forms_spec.py` for the long forms and re-run the render census; owner D, second E.

Actions agreed at Retrospective 2. All four are open actions carried into the next cycle; none is claimed as done. They sit in the feedback action log below so that the next sprint starts with them.

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5. Feedback action log

Consolidated log for both reviews, both retrospectives, and the two internal planning reviews held while writing `docs/07` – `docs/10`. **Source** distinguishes a review item from an internal one; an internal item is not presented as stakeholder feedback.

FB	Source	Issue raised	Agreed response	1st owner	2nd owner	Completion evidence	Status
FB-01	Sprint Review 1 (Q1)	The diagram carries 66 dashed message lines; the review asked whether that many are needed	Draw one representative message flow per pool pair, preferring exception hand-offs, and prove the paths still run	B	E	`docs/05-v2.1-changes.md` §1 (66 → 31; 421,424 → 174,820 px); `docs/03-test-record.md` §H3 (both paths clean)	**Complete**
FB-02	Sprint Review 1 (Q2)	Crossings rose from 42 to 148 and were not explained at the time	Confirm by A/B measurement, document it as an accepted trade, and defer a crossing-reduction pass	A	B	`docs/05-v2.1-changes.md` §4 A/B table; backlog PB-34	**Complete** (explained and accepted; not fixed)
FB-03	Sprint Review 1	Four flows clip a boundary event	Record the clipping; moving the boundary events would detach them from their host edge	B	A	`docs/05-v2.1-changes.md` §4 (named flows); `docs/03-test-record.md` §A4	**Not carried out as a fix.** Reason: moving the events costs more than the few clipped pixels. Outstanding impact is cosmetic, tracked as risk R-10 and test `TC-BPMN-04`
FB-04	Sprint Review 1 (Q3)	The Modeler reports 30 warnings	Analyse them, establish whether they are errors, and decide the closed-vs-open sub-process question with measurements	A	B	`docs/05-v2.1-changes.md` §4 (30 `no-bpmndi`, 0 errors; open variant measured at 21 lines through a shape vs 4); `screenshots/camunda-modeler-v2.1-status-bar.png`	**Complete**
FB-05	Sprint Review 1 (Q4)	The Tasklist Task pane is blank	Prove the forms with the renderer Tasklist uses, fix any real defect found, and treat the residual authentication failure as environmental	D	E	`docs/03-test-record.md` §F4 (DEF-12 and DEF-13 fixed; `TOTAL=36 FAILED=0`); `screenshots/camunda-forms-rendered.png`	**Partly complete**: form defects fixed; the `401` remains (FB-14, PB-32)

FB	Source	Issue raised	Agreed response	1st owner	2nd owner	Completion evidence	Status
FB-06	Sprint Review 1 (Q5)	Start events are 43 against a target of ≤ 2 per pool	Record the arithmetic and the protocol change required, and defer it; the message contract of a model that runs clean stays as it is	A	C	`docs/05-v2.1-changes.md` §4; backlog PB-33	**Not carried out.** Reason: a protocol change to a running model inside the timebox. Outstanding impact: the reduction target is unmet and risk R-10 stays open
FB-07	Retrospective 1 / Sprint Review 2 §3.3	Three modelled branches have never been driven end to end	Drive declined payment, confirmation lost and the three-month escalation early in Sprint 2	C	E	None recorded	**Not carried out**, because the committed compaction work and the preservation gate took the sprint. Outstanding impact: `TC-E2E-03` to `TC-E2E-05` remain `NOT RUN` and risk R-05 is open
FB-08	Retrospective 1	A drawing-only rebuild could silently drop a scored element	Build an executable preservation audit before any Sprint 2 drawing change	E	B	`tools/verify_preservation.py` ; `docs/03-test-record.md` §H1 (62/62)	**Complete**
FB-09	Sprint Review 2 (Q9)	`workers/README.md` documented 32 of the 35 model job types and its summary line was stale	Correct the table and the summary, or point it at the generated job-type list	C	E	`workers/README.md` : 36 rows (0–35), three types added, summary lines corrected, plus a one-line command to count the types from the model	**Done in R3.** DEF-14 and DEF-16 closed
FB-10	Sprint Review 2 (Q10)	The `zeebe:versionTag` is still 2.0.0 in the v2.1 artefact	Bump the tag for the next release and write the tag scheme into the configuration-management plan	E	A	Tag scheme documented in `docs/09-risk-contingency-and-config-management.md` §4.2	**Closed in v16.0.** The shipped file carries `versionTag` `16.0.0` on its single process, matching the release name; the tag scheme is in `docs/09` §4.2
FB-11	Sprint Review 2 (Q11) / internal planning review	`docs/06-visual-compaction.md` described a compaction (9 lanes, 12 sub-processes, 15 message flows) that was not in the shipped model	Treat the model as the source of truth and replace the document	B	A	`docs/06-diagram-engineering.md` (replaced); source-of-truth rule in `docs/07-product-backlog.md` A6 and `docs/09-...` §5 A7	**Done in R3**: defect DEF-15 closed

FB	Source	Issue raised	Agreed response	1st owner	2nd owner	Completion evidence	Status
FB-12	Internal planning review	There was no version-control repository in the workspace at the time of the review	Initialise a repository and commit the released state as the baseline before any further bulk regeneration	E	A	Repository created (`origin` = `github.com/Cczy1203/Advanced-information-systems-development`); tags `sr1-snapshot` , `sr2-snapshot` and `v14.0-final` ; scheme documented in `docs/09-risk-contingency-and-config-management.md` §4.1–4.2	**Closed.** Version control exists and is the version record. Note the separate, still-open point: the history shows the work arriving in three bulk commits, two after the submission date — that is risk R-14, not FB-12
FB-13	Retrospective 2 / internal planning review	Accessibility is partly delivered: no grouping headings and no WCAG check	Emit `group` components for the long forms, re-run the render census, and state the accessibility rationale	D	E	`docs/20-accessibility-and-the-forms.md` ; `screenshots/v16/accessibility-report.json` (37 forms, 237 fields, 15 groups, 0 findings); `screenshots/v16/tasklist-contrast.json`	**Partly closed on v16.0.** Grouping is delivered (15 `group` components) and every field carries a description (237/237). Still not met: no WCAG success-criterion mapping, contrast measured on one form rather than all 37, and no screen-reader or keyboard-only pass — `docs/20` states those limits rather than claiming conformance
FB-14	Internal planning review (defect DEF-17)	`docs/03-test-record.md` §F2 cites `screenshots/tasklist-camunda-form.png` , which does not exist	Re-capture the evidence once Tasklist authentication is fixed, or remove the reference and record the test as `NOT EVIDENCED`	E	D	`docs/08-test-plan.md` `TC-E2E-07` now records `NOT EVIDENCED` ; the source document is not yet corrected	**Partly complete**

5.1 Actions from the external Group Feedback sheet (R7, Group Mj)

The three rows below answer the three items written against this group in the external `Group Feedback` sheet. They are kept in the same eight-column form so that they can be read beside the internal rows, and they carry the same rule: a claim without an artefact behind it is not a completion.

FB	Source	Issue raised	Agreed response	1st owner	2nd owner	Completion evidence	Status
FB-15	External feedback, Group Feedback sheet **R7 item 1**	"Implement role-based access control in Camunda."	Declare the nine hospital roles and their members in the cluster configuration, map every user task's candidate group onto one of those roles, switch authorisation checking on, and prove the grants persist in the engine's own database rather than only in the file we wrote.	C	E	`bpm/camunda/application.yaml` (11 users, 10 declared roles, 20 role-authorisation blocks carrying 57 permission entries, `camunda.security.authorizations.enabled: true`); `bpm/tools/verify.ps1 -Check engine`; `bpm/screenshots/v15/rbac-enforcement.txt` §7 — USERS 11, ROLES 16, AUTHORIZATIONS 213, ROLE_MEMBER 11; USER_TASK grants 50 (our nine business roles × 4 permissions = 36, plus 14 built in); our ten roles hold 57 grants; deployment key 2251799813685383 carrying 9 process definitions and 36 forms on Camunda 8.10.0-alpha5	**Complete** for the engine layer. Two configuration traps were found by experiment and are now written into the file as comments: a grant must state `resourceId` explicitly (a wildcard is `resourceId: "*"`), and a user-task permission must use the short name `READ` / `CLAIM` / `COMPLETE` / `UPDATE` — the long forms start the engine without complaint but silently persist no USER_TASK grant at all. The API stays `unprotected-api: true`; the reason is recorded under limitation L-1 in `docs/14-feedback-response.md`
FB-16	External feedback, Group Feedback sheet **R7 item 2**	"Improve form validation, particularly for date selections, to prevent invalid or inappropriate dates."	Establish from the renderer's own source what a form can and cannot enforce, then split the work along that line: the form carries what it can express, and a shared rule table drives both the forms and a worker-side validator for what it cannot.	D	C	`bpm/tools/forms-inventory-v14.json` → `bpm/tools/forms-inventory-v15.json` measured with one ruler: required fields 142 → **147**, `pattern` rules 0 → **63**, outstanding findings 6 → **0**; `disallowPassedDates` added where a date is a future commitment; `bpm/workers/src/main/resources/date-rules.json` as the single source of truth; `bpm/workers/.../support/DateRuleValidator.java` + `DateRuleValidatorTest` (14 cases, all passing)	**Complete.** The decisive finding is that form-js implements `disallowPassedDates` as a flatpickr `minDate`, so it constrains the calendar widget only and never runs in the renderer's validation function; a date outside the rule can still be submitted straight to the API. That is why the worker-side check is a required second layer and not a duplicate of the form

FB	Source	Issue raised	Agreed response	1st owner	2nd owner	Completion evidence	Status
FB-17	External feedback, Group Feedback sheet **R7 item 3**	"Implement database integration to support data persistence."	Put the state that used to live in process memory into an embedded H2 database: a versioned schema and idempotent seed, a connection pool, a repository that keeps the existing public API so the 36 workers are untouched, reference data the process can be validated against, and a demonstration that the data survives a cold restart.	C	B	`com.h2database:h2:2.2.224` in `bpm/workers/pom.xml`; `bpm/workers/src/main/resources/db/schema.sql` (15 tables) and `db/seed.sql` (3 practices, 3 patients, 11 roles, 63 grants, `MERGE`-based and re-runnable); `bpm/workers/.../db/HospitalDatabase.java`, `ReferenceData.java`, `AccessControl.java`, `NhsNumber.java`; `support/HospitalStore.java` rewritten on JDBC with the v14 signature unchanged; `bpm/tools/verify.ps1 -Check persistence`; `bpm/screenshots/v15/h2-persistence.txt`	**Complete.** Evidence of real persistence: the first cold run reports `appointment=0`; a row is inserted by hand; the **second cold run of the same jar** reports `appointment=1`, and `patient=3 / gp_practice=3 / role_permission=63` are identical in both runs, which also shows the seed is idempotent. 30 worker tests pass, of which 5 assert exactly this across a close-and-reopen

Rule applied to this log. An action that was not carried out keeps a row, a stated reason and a described outstanding impact. Nothing is dropped because it was inconvenient; FB-03, FB-06, FB-07 and the open items above are the honest remainder of two sprints.

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6. Release notes

6.1 Initial Release (R1): v2.0

Model and forms. One BPMN collaboration: 15 participants (9 executable processes, 6 black-box external bodies), 14 lanes, 301 flow nodes, 311 sequence flows, 31 message flows, 39 user tasks, 47 service tasks across 35 job types, 41 gateways, 18 boundary events, 9 named error codes and 3 collapsed sub-processes. It ships 36 Camunda Forms.

Delivered backlog items.

Backlog items	Release note
PB-01 – PB-04	Measured gap audit; the case-study collaboration rebuilt as one file with correlated message hand-offs and lanes where a pool covers several desks
PB-05, PB-06	New layout engine: zero sequence-flow overlaps, zero message-flow overlaps, zero diagonals; repetitive administration contained in three collapsed sub-processes

Backlog items	Release note
PB-07, PB-08, PB-19	Every modelled exception has an owner and an exit; clinic-letter timers at 7 days and weekly thereafter with a three-rung escalation ladder; external capacity has a counted retry capped at 3
PB-18	Compensation added for the provisional appointment series and the cycle booking (two boundary events, two handlers, two throws), modelled but not triggered in a live run
PB-09 – PB-11	A form on every user task (39/39), a candidate group on every user task (39/39), and required/length/range validation
PB-13 – PB-17	36 worker classes subscribing to 37 job-type names; simulated external services with documented forced-failure switches; duplicate suppression; no card data; non-overwriting audit workers
PB-20 – PB-23	One-command deployment of 9 process definitions and 36 forms; both scripted paths complete with no incidents; all 36 forms render in form-js
PB-24	Test record with expected vs actual for every check, and a modelling-decisions document with numbered assumptions and limitations

Known issues in R1. PB-12 partly delivered (descriptions only, no grouping); compensation not triggered live; three branches not run; Tasklist cannot draw the forms in this c8run setup; 30 Modeler warnings, all `no–bpmndi` inside collapsed sub-processes; four flows clip a boundary event by a few pixels; crossings 147; idempotency state in memory.

6.2 Second Release (R2): v2.1

Change summary. Drawing-only release: message flows 66 → 31 (dashed line 421,424 → 174,820 px, –59%); routine end events 60 → 32 with exception ends never merged; lanes 20 → 14. No scored element removed, proved by 62 parsed preservation checks.

Delivered backlog items. PB-25 (preservation audit), PB-26 (message-flow thinning), PB-27 (end-event merge), PB-28 (lane consolidation), PB-29 (Modeler re-validation, diagram and section export).

Verification. 62/62 preservation checks; Camunda Modeler 5.51.0 on Camunda 8.10 (alpha) reporting 0 errors; `validate_model.js` reporting 0 warnings, 0 unresolved references, 0 dangling flows and full form/candidate-group/job-type coverage; both paths re-run with `incidents: none`; deployed as processes v17 and forms v18; Operate showing 21 running instances with 0 incidents.

Known issues in R2. Crossings rose to 197 as the measured cost of the end-event merge; 43 message start events remain against a target of ≤ 2 per pool; the four clipped lines remain; sub-process boxes stay collapsed (30 `no–bpmndi` warnings); the three undriven branches and the live compensation trigger remain undemonstrated; `versionTag` still reads 2.0.0; documentation drift DEF-14 to DEF-17 is recorded but not corrected.

Carried into the next cycle. PB-30, PB-31, PB-32, PB-33, plus the open feedback actions FB-09 to FB-14. PB-34 and PB-35 were taken up in R3.

6.3 Third Release (R3): v7.0

Change summary. Two modelling changes and a routing change, all aimed at the same thing: a drawing a reader can follow.

	v2.1	v7.0
End events	32, several merged	**65, none with more than one incoming flow**
Drawn message flows	31	**15** (11 exception hand-offs + 4 system-boundary hand-offs)

	v2.1	v7.0
Sequence-flow crossings	197	**94**
Lines drawn through a shape	4	**0**
Total ink	378,475 px	**182,317 px**
Canvas	7,496 × 13,736	**5,332 × 14,176**
`bpmnlint:recommended` errors	39	**2**
Preservation checks	61 of 62	**67 of 67**

Delivered backlog items. PB-34 (crossing reduction, 197 → 94, clipped lines 4 → 0) and the new PB-35 (one end event per terminating path; dashed lines cut to the hand-offs that carry an outcome). Also closed: DEF-14, DEF-15 and DEF-16 from `docs/08-test-plan.md` §9. The drifted diagram document was replaced, the worker job-type catalogue completed, and the collapsed sub-process steps given diagram interchange so the boxes open correctly in the Modeler.

Known issues in R3. The two `no-implicit-start` errors (DEF-18) remain: a withdrawn v2.0 retry chain and one step inside a collapsed sub-process. Both are unreachable code on a model that runs clean; repairing them is a message-protocol change and was deliberately not made. 43 message start events still exceed the 2-per-pool target, for the reason recorded in `docs/05-v2.1-changes.md` §4. DEF-17 (one cited screenshot that does not exist) is still open because the Tasklist cannot be re-authenticated to recapture it.

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7. Assumptions

1. **A1: Dates and attendees.** No sprint dates, review dates, attendee lists or minutes exist in the workspace. This document records the substance of each review (goal, increment, evidence, decisions, questions and limitations) without inventing a calendar or a quorum.
2. **A2: Graded stand-ups.** Each sprint ran a graded stand-up and a graded Sprint Review. Stand-up output is retained as a separate artefact — see 19-standups.md, which records both stand-ups with the evidence each update points at. (The earlier note here said it was not retained; that was true when it was written and is corrected now.); the blockers it produced are the FB rows in §5 whose source is a retrospective or an internal review.
3. **A3: Quotation.** Questions in §1.5 and §3.5 are paraphrases reconstructed from the decisions and changes recorded in `docs/05-v2.1-changes.md`, `docs/02-status.md` and `docs/03-test-record.md`. They are not verbatim transcriptions.
4. **A4: Internal items.** FB-11 to FB-14 come from the internal planning review performed while writing `docs/07` – `docs/10`, not from a graded review. They are labelled as such in the Source column so they are not presented as stakeholder feedback.
5. **A5: Retrospective 1 timing.** The improvement action to build the preservation gate before Sprint 2 is evidenced by the existence and first use of `tools/verify_preservation.py` in `docs/03-test-record.md` §H1; no retrospective minute records it.
6. **A6: Release scope.** The Initial Release is R1 (v2.0), the second release is R2 (v2.1) and the third is R3 (v7.0), per `docs/07-product-backlog.md` §2 and Assumption A8 there.

R5 (v16.0) — the final review of the shipped release

The two reviews above are the v2.0 / v2.1 / v7.0 record. This section is what changed at v16.0, so a reader can see the shipped position without reading the whole history.

Increment presented: `model/UFCEP6-0-3_Hospital_Patient_Pathway_v16.bpmn` — one hospital pool, three responsibility lanes holding nine team lanes, six black-box outside participants. Deployed to c8run 8.10.0-alpha5 and driven by `tools/run_v16.py`.

Against the sprint goal: met for the pathway itself. Eight scenarios COMPLETED (`docs/03` section K2), including the compensation handler, which v14 had claimed and v16 had to add.

Decisions taken:

- the hospital's own desks are lanes, not pools — `docs/17` section 1
- compensation is modelled where the business needs it, not everywhere
- the implicit convergences are left and counted rather than excused — `docs/17` §4

Acceptance evidence: `screenshots/run-v16-*.log` , `screenshots/v16/engine-record.txt` , `screenshots/v16/accessibility-report.json` .

Not finished, and carried into the next increment:

- the worker failure paths on v16 (`docs/03` K3, **NOT RUN**)
- worker-layer unit tests (PB-45)
- a recorded demonstration with a person using Tasklist (PB-46)
- the silent fall-back to a non-persistent store when two workers share the H2 file (K5.4) — a code defect, not an operational one, and not fixed

Feedback that drove this increment: an audit of the submission, which found the v16 model absent from git, a regression of DEF-02 in which every consenting patient was routed to the decline branch, two results contradicted by their own cited evidence, and a shipped jar built from an earlier release. All four are closed; the closure is in `docs/17` sections 5 and 6 and `docs/03` section K5.